

Thesis progress report

LLM-Guided Neural Network Compilation for Real-time UAV Edge Inference

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A tiny network proves the rewrite

The planner is not allowed to edit the graph. This report checks that the engine, which is allowed to edit it, really does.

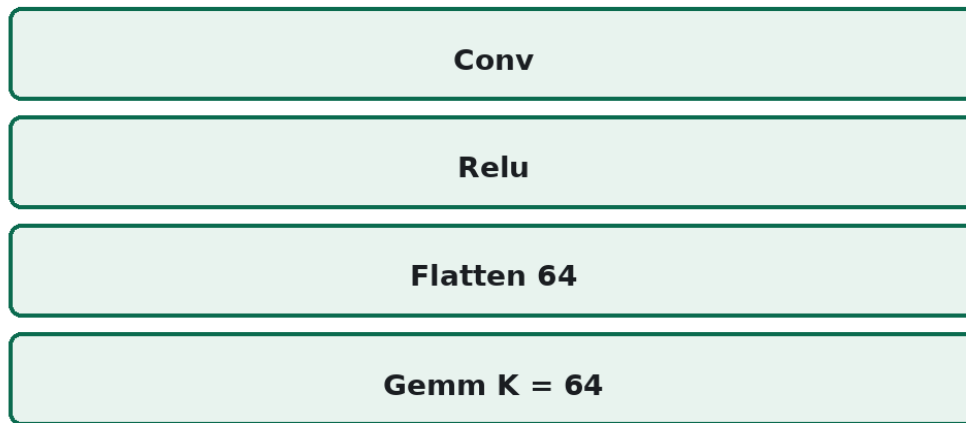
The network used here is small enough to read by hand. Its input is one map of size $1 \times 1 \times 8 \times 8$. A convolution keeps a spatial map, a ReLU follows it, a pooling layer shrinks the map, Flatten turns the result into a vector of 64 values, and a Gemm produces the output. The Gemm width has to be 64, because that is the Flatten width. A width of 16 was an early mistake on this machine. It is kept as a test that must fail, so a broken contract cannot be reported as a successful compile.

The pass used is `graph_fuse`. It joins a convolution with an immediately following ReLU. On ONNX Runtime for the CPU, the ReLU node leaves the graph and a `FusedConv` node takes its place. The Flatten width is unchanged. The proof is the operator list before and after the pass, not a log line that merely says the pass ran.

Tests lock both facts. The broken width must not compile as a valid model. `graph_fuse` on the correct fixture must leave one `FusedConv` and no ReLU. That is the first evidence that the rewrite is a real graph change.

The width of 16 came from a real mistake, not from a made-up test. The first reading was that Gemm K should match the channel count, or a 4 by 4 spatial guess. The Flatten after pooling is 4 by 4 by 4, so K is 64. The session compiled only after that was corrected. Keeping the broken file means the same wrong guess cannot pass quietly later. A log line that says fusion ran was also not enough. The operator list has to change, or the pass is only a name.

Before



After graph_fuse

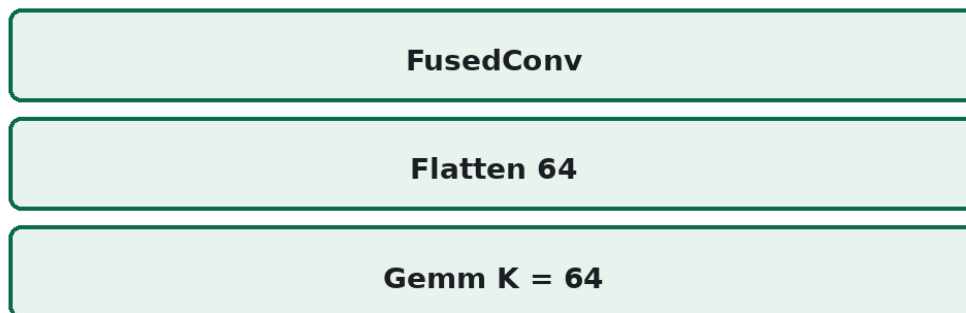


Figure 1. Conv followed by ReLU becomes one FusedConv. Flatten stays at 64 values.

The rewrite is visible on this small graph. The same commands still have to be shown on the networks a companion computer would actually run.